

C950 Task-2 WGUPS Write-Up

C950 Task-2 WGUPS Write-Up

(Task-2: The implementation phase of the WGUPS Routing Program).

(Zip your source code and upload it with this file)

Kalvin Patarawong

Student ID:

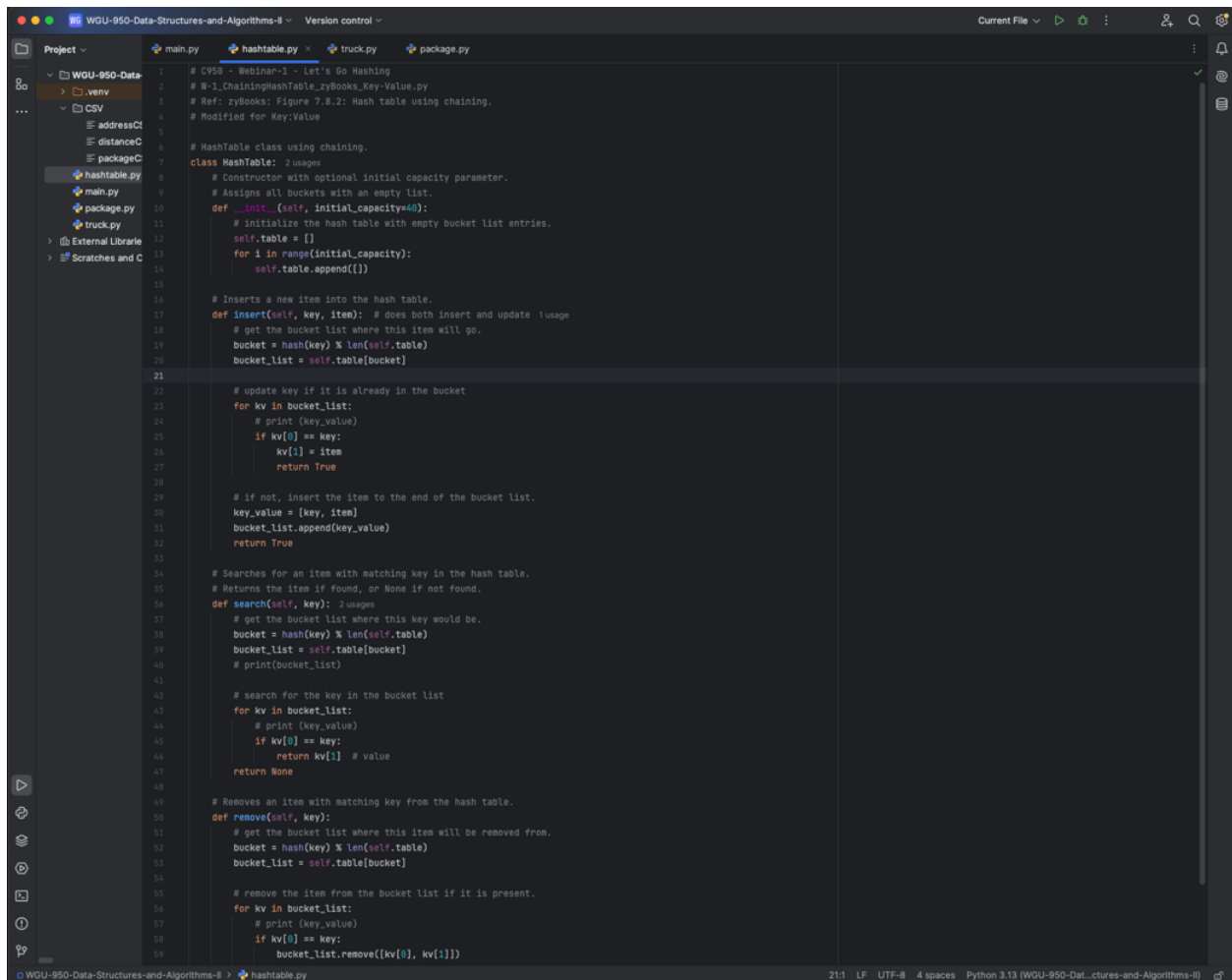
WGU Email:

Date: 10/7/2024

C950 Data Structures and Algorithms II

A. Hash Table

Hash Table Screenshot goes here



```
1 # C950 - Webinar-1 - Let's Go Hashing
2 # W-1.ChainingHashTable.zyBooks.Key-Value.py
3 # Ref: zyBooks: Figure 7.8.2: Hash table using chaining.
4 # Modified for Key:Value
5
6 # HashTable class using chaining.
7 class HashTable:
8     # Constructor with optional initial capacity parameter.
9     # Assigns all buckets with an empty list.
10     def __init__(self, initial_capacity=40):
11         # initialize the hash table with empty bucket list entries.
12         self.table = []
13         for i in range(initial_capacity):
14             self.table.append([])
15
16     # Inserts a new item into the hash table.
17     def insert(self, key, item): # does both insert and update
18         # get the bucket list where this item will go.
19         bucket = hash(key) % len(self.table)
20         bucket_list = self.table[bucket]
21
22         # update key if it is already in the bucket
23         for kv in bucket_list:
24             # print(key_value)
25             if kv[0] == key:
26                 kv[1] = item
27                 return True
28
29         # if not, insert the item to the end of the bucket list.
30         key_value = [key, item]
31         bucket_list.append(key_value)
32         return True
33
34     # Searches for an item with matching key in the hash table.
35     # Returns the item if found, or None if not found.
36     def search(self, key):
37         # get the bucket list where this key would be.
38         bucket = hash(key) % len(self.table)
39         bucket_list = self.table[bucket]
40         # print(bucket_list)
41
42         # search for the key in the bucket list
43         for kv in bucket_list:
44             # print(key_value)
45             if kv[0] == key:
46                 return kv[1] # value
47         return None
48
49     # Removes an item with matching key from the hash table.
50     def remove(self, key):
51         # get the bucket list where this item will be removed from.
52         bucket = hash(key) % len(self.table)
53         bucket_list = self.table[bucket]
54
55         # remove the item from the bucket list if it is present.
56         for kv in bucket_list:
57             # print(key_value)
58             if kv[0] == key:
59                 bucket_list.remove(kv[0], kv[1])
```

B. Look-Up Functions

Look-up function screenshot goes here

```
48
49 # Load distance data between two locations
50 def loadDistanceData(x_value, y_value): 1 usage
51     distance = DistanceCSV[x_value][y_value]
52     if distance == '':
53         distance = DistanceCSV[y_value][x_value]
54     return float(distance)
55
56 # Load address data for matching delivery location
57 def loadAddressData(address): 2 usages
58     for row in AddressCSV:
59         if address in row[2]:
60             return int(row[0])
61     return None
62
63
```

C. Original Code

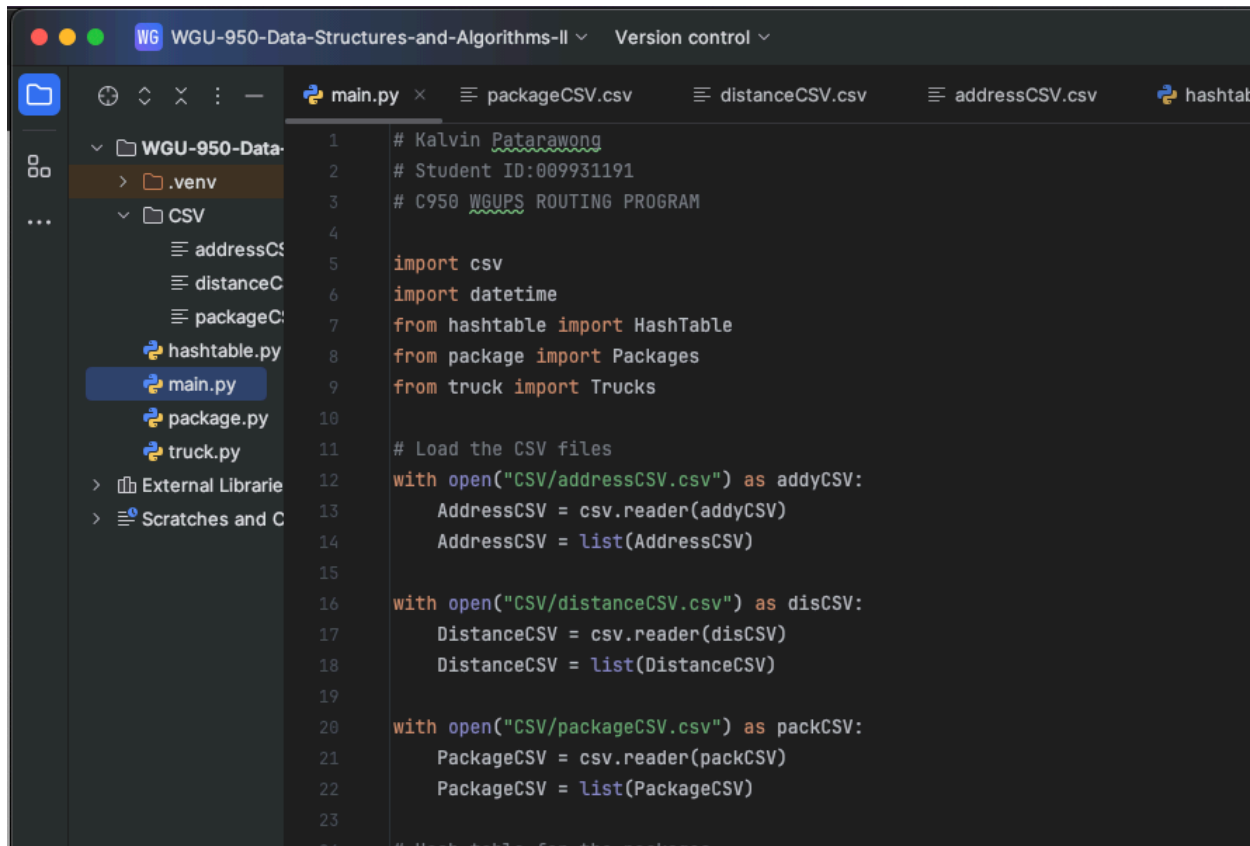
Major code blocks screenshots go here showing implementation

```

101 # Main.py
102 # packageCSV.csv
103 # distanceCSV.csv
104 # addressCSV.csv
105 # hashtable.py
106 # truck.py
107 # package.py
108
109 # Nearest neighbor algorithm to deliver the packages on truck
110 def truckDeliverPackages(truck):
111     # 3 usages
112     # List for all packages that needs to be delivered
113     enroute = []
114     # Put packages from the hash table into list
115     for packageID in truck.packages:
116         package = packageHash.search(packageID)
117         if package: # Only add if package is found
118             enroute.append(package)
119
120     truck.packages.clear()
121
122     # While there are packages left to be delivered, the algorithm will run
123     while len(enroute) > 0:
124         nextAddy = float('inf') # Set an initial very high value
125         nextPackage = None
126
127         for package in enroute:
128             address_id = loadAddressData(package.street)
129             if address_id is None:
130                 continue
131
132             distance = loadDistanceData(loadAddressData(truck.currentLocation), address_id)
133             if distance <= nextAddy:
134                 nextAddy = distance
135                 nextPackage = package
136
137         if nextPackage:
138             truck.packages.append(nextPackage.ID)
139             enroute.remove(nextPackage)
140             truck.miles += nextAddy
141             truck.currentLocation = nextPackage.street
142             truck.time += datetime.timedelta(hours=nextAddy / 18) # Time = distance / speed (converted to hours)
143             nextPackage.deliveryTime = truck.time
144             nextPackage.departureTime = truck.departTime
145
146 # Load package CSV data and deliver packages
147 loadPackageData('CSV/packageCSV.csv')
148
149 # Manually load truck and assign packages to departure time
150 truck1 = Trucks(speed=18, miles=0.0, currentLocation="4001 South 700 East", datetime.timedelta(hours=0), packages=[1,13,14,15,16,19,20,27,29,30,31,34,37,40])
151 truck2 = Trucks(speed=18, miles=0.0, currentLocation="4001 South 700 East", datetime.timedelta(hours=11), packages=[2,3,4,5,9,18,26,28,32,35,36,38])
152 truck3 = Trucks(speed=18, miles=0.0, currentLocation="4001 South 700 East", datetime.timedelta(hours=9, minutes=5), packages=[6,7,8,10,11,12,17,21,22,23,24,25,33,39])
153
154 # Put trucks through loading process
155 truckDeliverPackages(truck1)
156 truckDeliverPackages(truck2)
157
158 # The below line of code ensures that truck 3 does not leave until first two trucks finish delivering packages
159 truck3.departTime = min(truck1.time, truck2.time)
160 truckDeliverPackages(truck3)
161
162 # Print title
163 print("Western Governors University Parcel Service (WGUPS)")
164
165 # User Interface
166 def show_menu():
167     print("\n*****")
168
169 # Main.py
  
```

C1. Identification Information

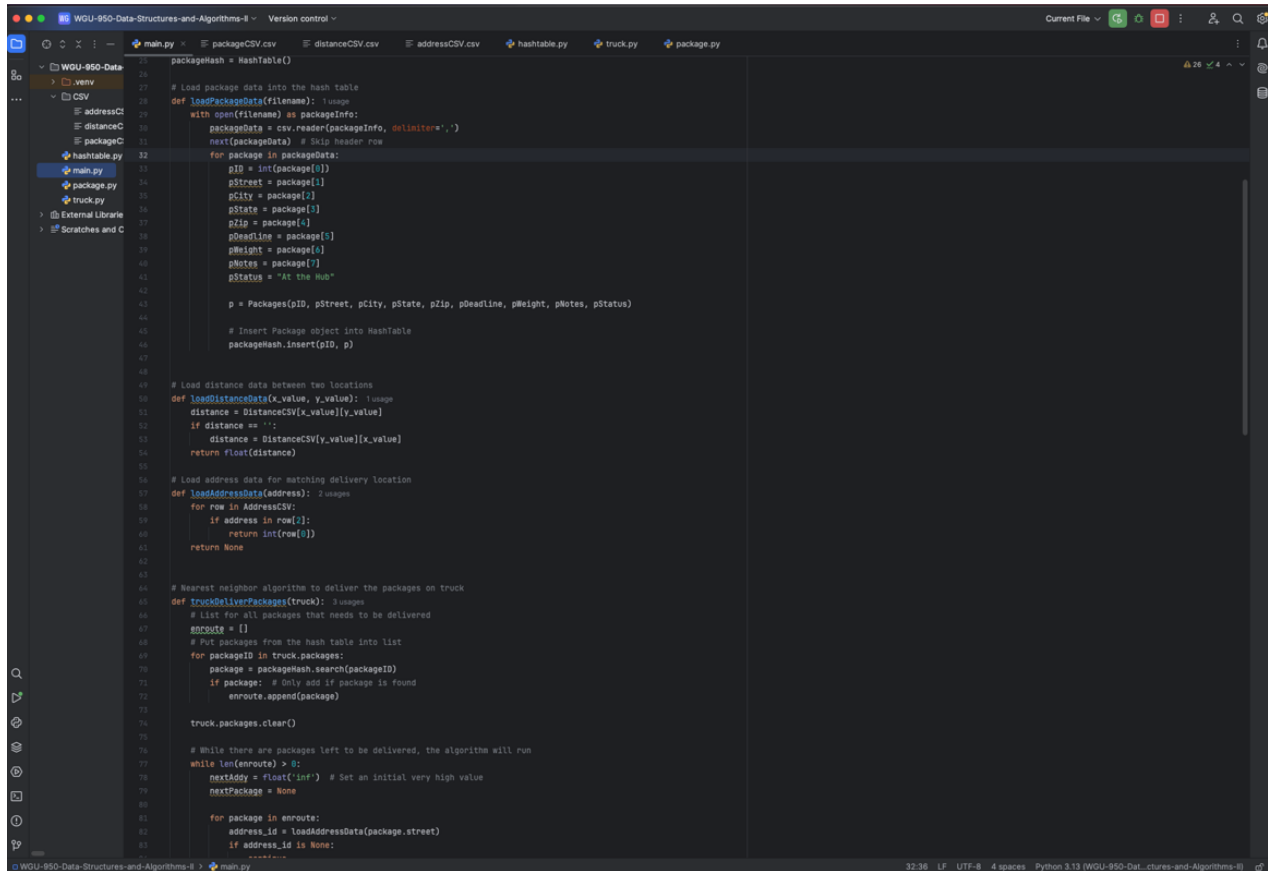
main.py screenshot goes here showing Student ID



```
1 # Kalvin Patarawong
2 # Student ID:009931191
3 # C950 WGUPS ROUTING PROGRAM
4
5 import csv
6 import datetime
7 from hashtable import HashTable
8 from package import Packages
9 from truck import Trucks
10
11 # Load the CSV files
12 with open("CSV/addressCSV.csv") as addyCSV:
13     AddressCSV = csv.reader(addyCSV)
14     AddressCSV = list(AddressCSV)
15
16 with open("CSV/distanceCSV.csv") as disCSV:
17     DistanceCSV = csv.reader(disCSV)
18     DistanceCSV = list(DistanceCSV)
19
20 with open("CSV/packageCSV.csv") as packCSV:
21     PackageCSV = csv.reader(packCSV)
22     PackageCSV = list(PackageCSV)
23
24 # Hash table for the packages
```

C2. Process and Flow Comments

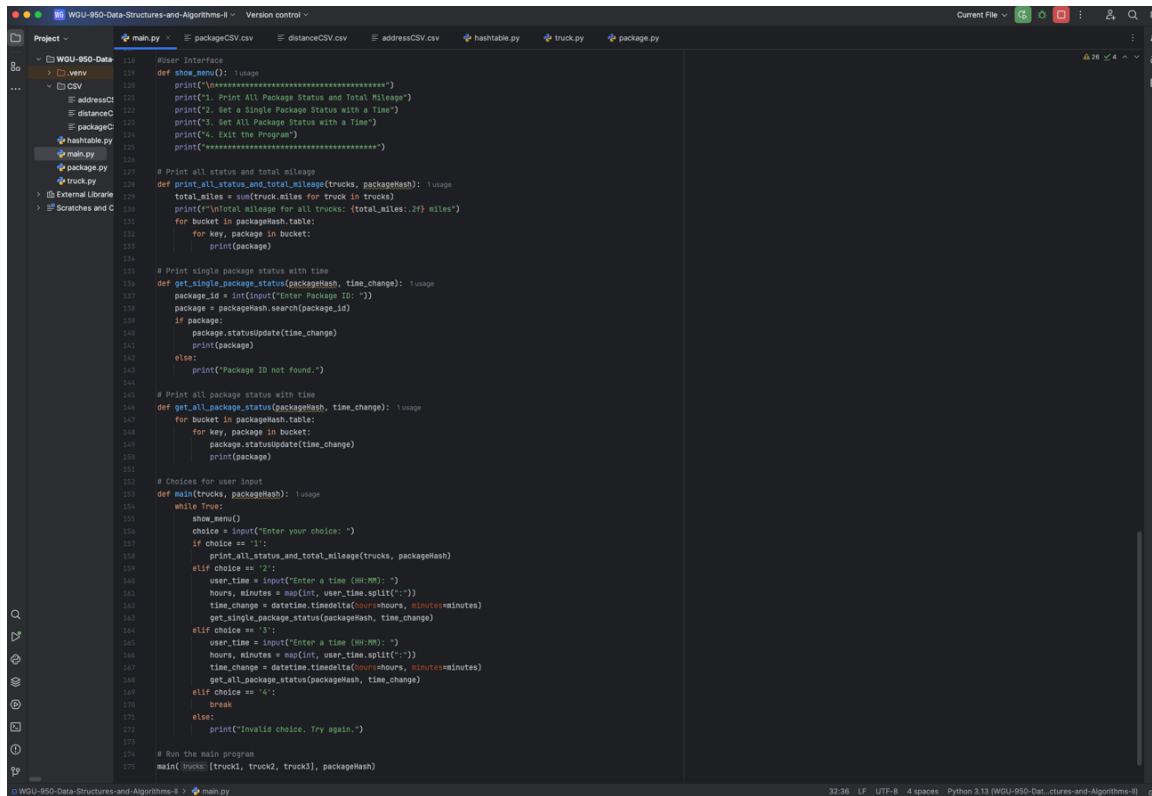
Some code blocks screenshots go here showing comments



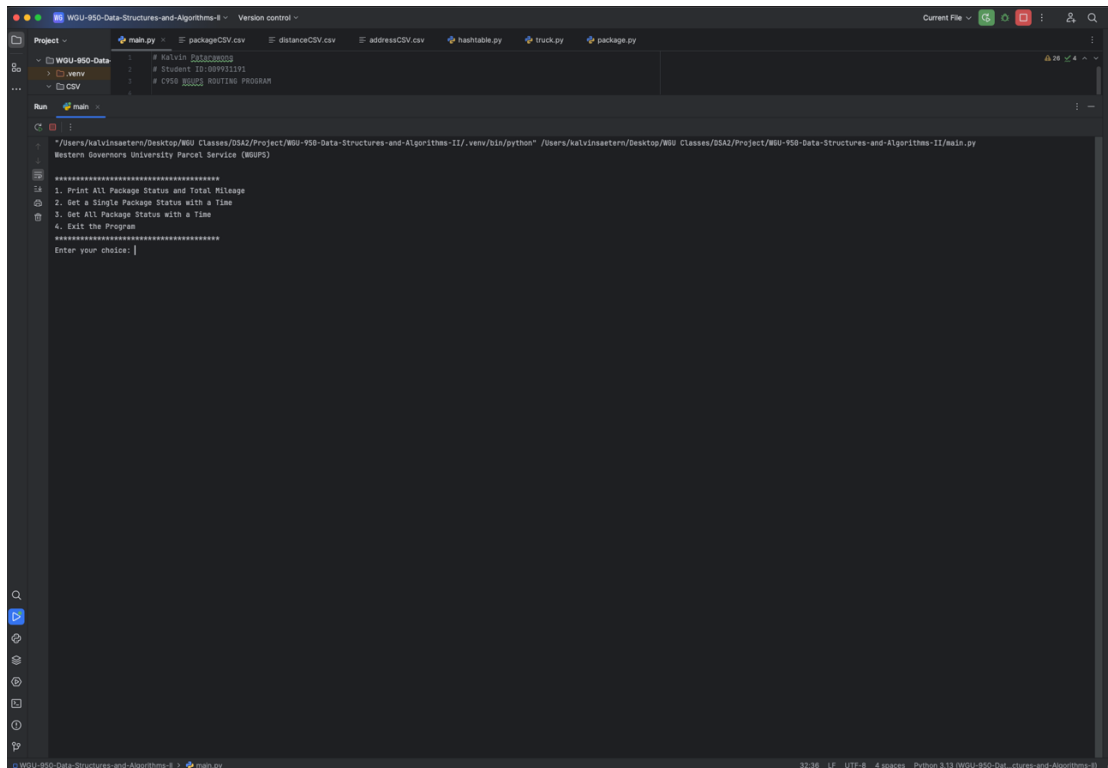
```
21 packageHash = HashTable()
22
23 # Load package data into the hash table
24 def loadPackageData(filename):
25     with open(filename) as packageInfo:
26         packageData = csv.reader(packageInfo, delimiter=',')
27         next(packageData) # Skip header row
28         for package in packageData:
29             pID = int(package[0])
30             pStreet = package[1]
31             pCity = package[2]
32             pState = package[3]
33             pZip = package[4]
34             pHeadline = package[5]
35             pWeight = package[6]
36             pNotes = package[7]
37             pStatus = "At the Hub"
38
39             p = Packages(pID, pStreet, pCity, pState, pZip, pHeadline, pWeight, pNotes, pStatus)
40
41             # Insert Package object into HashTable
42             packageHash.insert(pID, p)
43
44 # Load distance data between two locations
45 def loadDistanceData(x_value, y_value):
46     distance = DistanceCSV[x_value][y_value]
47     if distance == "":
48         distance = DistanceCSV[y_value][x_value]
49     return float(distance)
50
51 # Load address data for matching delivery location
52 def loadAddressData(address):
53     for row in AddressCSV:
54         if address in row[2]:
55             return int(row[0])
56     return None
57
58 # Nearest neighbor algorithm to deliver the packages on truck
59 def TruckDeliverPackages(truck):
60     # List for all packages that needs to be delivered
61     enroute = []
62     # Put packages from the hash table into list
63     for packageID in truck.packages:
64         package = packageHash.search(packageID)
65         if package: # Only add if package is found
66             enroute.append(package)
67
68     truck.packages.clear()
69
70 # While there are packages left to be delivered, the algorithm will run
71 while len(enroute) > 0:
72     nextAddy = float('inf') # Set an initial very high value
73     nextPackage = None
74
75     for package in enroute:
76         address_id = loadAddressData(package.street)
77         if address_id is None:
```

D. Interface

Interface screenshot goes here



```
118 # User Interface
119 def show_menu(): usage
120     print("\n=====")
121     print("1. Print All Package Status and Total Mileage")
122     print("2. Get a Single Package Status with a Time")
123     print("3. Get All Package Status with a Time")
124     print("4. Exit the Program")
125     print("=====")
126
127 # Print all status and total mileage
128 def print_all_status_and_total_mileage(trucks, packageHash): usage
129     total_miles = sum(truck.miles for truck in trucks)
130     print(f"Total mileage for all trucks: {total_miles:.2f} miles")
131     for bucket in packageHash.table:
132         for key, package in bucket:
133             print(package)
134
135 # Print single package status with time
136 def get_single_package_status(packageHash, time_change): usage
137     package_id = int(input("Enter Package ID: "))
138     package = packageHash.search(package_id)
139     if package:
140         package.statusUpdate(time_change)
141         print(package)
142     else:
143         print("Package ID not found.")
144
145 # Print all package status with time
146 def get_all_package_status(packageHash, time_change): usage
147     for bucket in packageHash.table:
148         for key, package in bucket:
149             package.statusUpdate(time_change)
150             print(package)
151
152 # Choices for user input
153 def main(trucks, packageHash): usage
154     while True:
155         show_menu()
156         choice = input("Enter your choice: ")
157         if choice == "1":
158             print_all_status_and_total_mileage(trucks, packageHash)
159         elif choice == "2":
160             user_time = input("Enter a time (HH:MM): ")
161             hours, minutes = map(int, user_time.split(":"))
162             time_change = datetime.timedelta(hours=hours, minutes=minutes)
163             get_single_package_status(packageHash, time_change)
164         elif choice == "3":
165             user_time = input("Enter a time (HH:MM): ")
166             hours, minutes = map(int, user_time.split(":"))
167             time_change = datetime.timedelta(hours=hours, minutes=minutes)
168             get_all_package_status(packageHash, time_change)
169         elif choice == "4":
170             break
171         else:
172             print("Invalid choice. Try again.")
173
174 # Run the main program
175 main([truck1, truck2, truck3], packageHash)
```



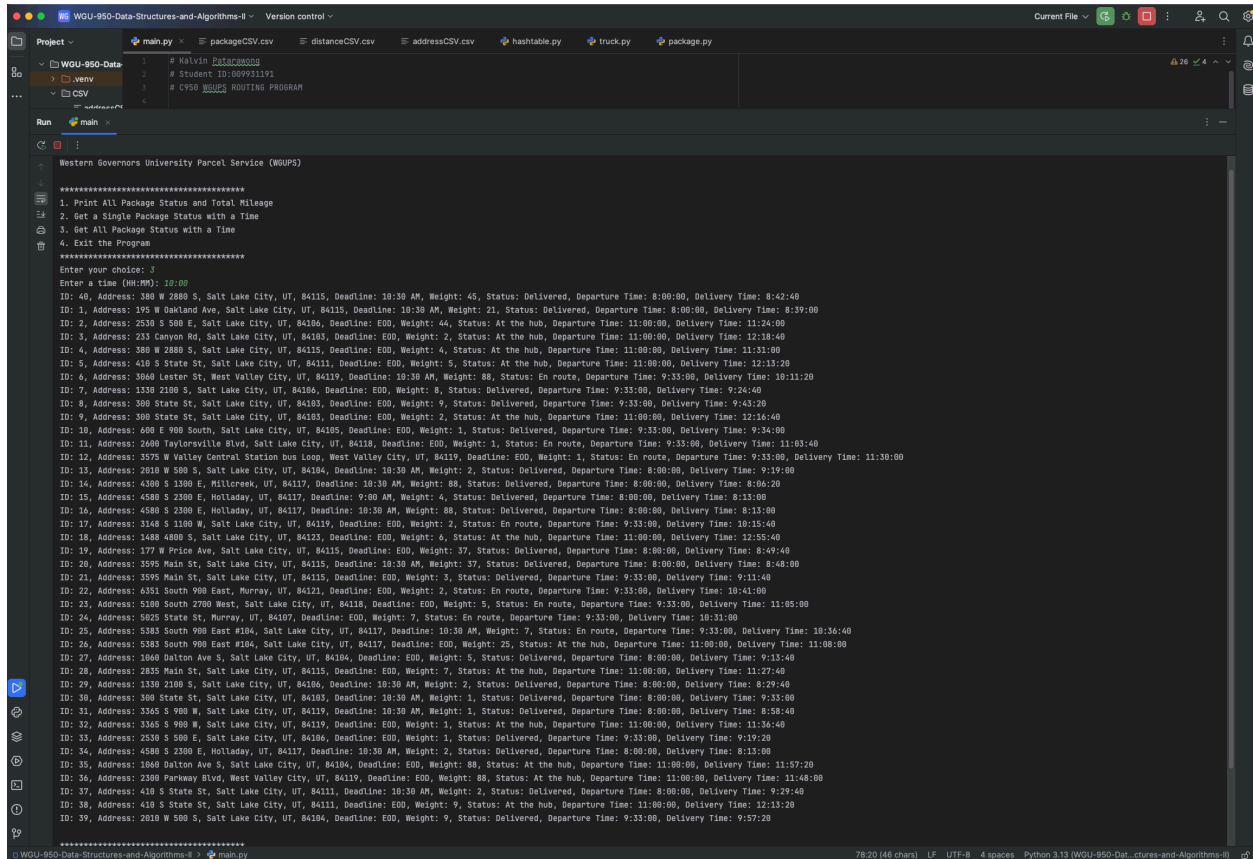
```
1 # Halvin Package0003
2 # Student ID:089931191
3 # C950 8000 ROUTING PROGRAM
4
Run main
/Users/halvinaeston/Desktop/WGU Classes/DSAI/Project/WGU-950-Data-Structures-and-Algorithms-II/venv/bin/python /Users/halvinaeston/Desktop/WGU Classes/DSAI/Project/WGU-950-Data-Structures-and-Algorithms-II/main.py
Western Governors University Parcel Service (WGUPS)
=====
1. Print All Package Status and Total Mileage
2. Get a Single Package Status with a Time
3. Get All Package Status with a Time
4. Exit the Program
=====
Enter your choice: |
```

Screenshot goes here

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D2. Second Status Check

Screenshot goes here



```
WGUPS-Data-Structures-and-Algorithms-II - Version control
Project - main.py packageCSV.csv distanceCSV.csv addressCSV.csv hashtable.py truck.py package.py
WGUPS-950-Data-Structures-and-Algorithms-II
main.py
# Kevin Datta
# Student ID: 00991191
# C950 WGUPS ROUTING PROGRAM

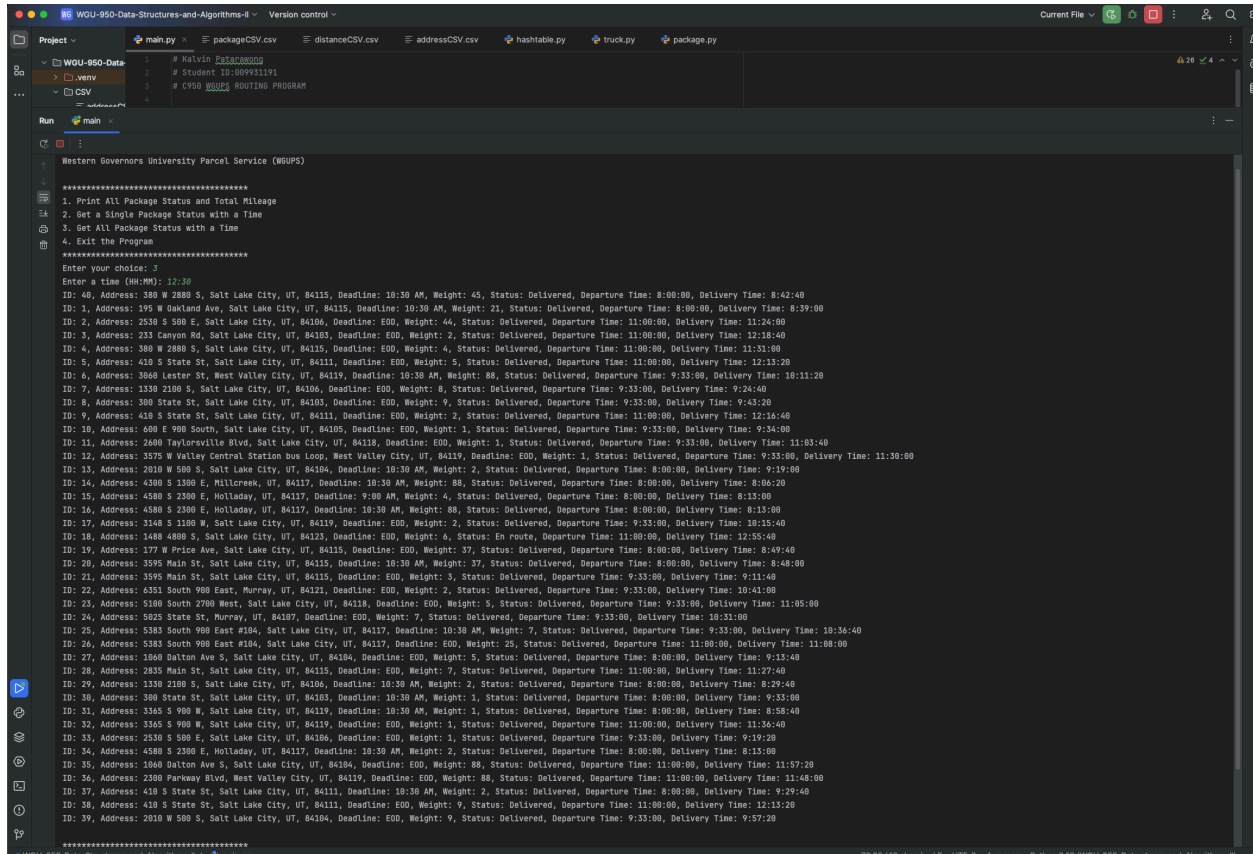
Run main
Western Governors University Parcel Service (WGUPS)

*****
1. Print All Package Status and Total Mileage
2. Get a Single Package Status with a Time
3. Get All Package Status with a Time
4. Exit the Program
*****
Enter your choice: 1
Enter a time (HH:MM): 10:00
ID: 40, Address: 380 W 2880 S, Salt Lake City, UT, 84115, Deadline: 10:30 AM, Weight: 45, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 8:42:40
ID: 1, Address: 195 W Oakland Ave, Salt Lake City, UT, 84115, Deadline: 10:30 AM, Weight: 21, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 8:39:00
ID: 2, Address: 2530 S 500 E, Salt Lake City, UT, 84106, Deadline: EOD, Weight: 44, Status: At the hub, Departure Time: 11:00:00, Delivery Time: 11:24:00
ID: 3, Address: 233 Canyon Rd, Salt Lake City, UT, 84103, Deadline: EOD, Weight: 2, Status: At the hub, Departure Time: 11:00:00, Delivery Time: 12:18:40
ID: 4, Address: 380 W 2880 S, Salt Lake City, UT, 84115, Deadline: EOD, Weight: 4, Status: At the hub, Departure Time: 11:00:00, Delivery Time: 11:31:00
ID: 5, Address: 418 S State St, Salt Lake City, UT, 84111, Deadline: EOD, Weight: 5, Status: At the hub, Departure Time: 11:00:00, Delivery Time: 12:13:20
ID: 6, Address: 5060 Lester St, West Valley City, UT, 84119, Deadline: 10:30 AM, Weight: 88, Status: En route, Departure Time: 9:33:00, Delivery Time: 10:11:20
ID: 7, Address: 1330 2100 S, Salt Lake City, UT, 84106, Deadline: EOD, Weight: 8, Status: Delivered, Departure Time: 9:33:00, Delivery Time: 9:24:40
ID: 8, Address: 300 State St, Salt Lake City, UT, 84103, Deadline: EOD, Weight: 9, Status: Delivered, Departure Time: 9:33:00, Delivery Time: 9:43:20
ID: 9, Address: 300 State St, Salt Lake City, UT, 84103, Deadline: EOD, Weight: 2, Status: At the hub, Departure Time: 11:00:00, Delivery Time: 12:16:40
ID: 10, Address: 400 E 900 South, Salt Lake City, UT, 84105, Deadline: EOD, Weight: 1, Status: Delivered, Departure Time: 9:33:00, Delivery Time: 9:34:00
ID: 11, Address: 2600 Taylorsville Blvd, Salt Lake City, UT, 84119, Deadline: EOD, Weight: 1, Status: En route, Departure Time: 9:33:00, Delivery Time: 11:03:40
ID: 12, Address: 3575 W Valley Central Station Bus Loop, West Valley City, UT, 84119, Deadline: EOD, Weight: 1, Status: En route, Departure Time: 9:33:00, Delivery Time: 11:30:00
ID: 13, Address: 2610 W 500 S, Salt Lake City, UT, 84104, Deadline: 10:30 AM, Weight: 2, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 9:19:00
ID: 14, Address: 4300 S 1300 E, Millcreek, UT, 84117, Deadline: 10:30 AM, Weight: 80, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 8:04:20
ID: 15, Address: 4500 S 2300 E, Holladay, UT, 84117, Deadline: 9:00 AM, Weight: 4, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 8:13:00
ID: 16, Address: 4500 S 2300 E, Holladay, UT, 84117, Deadline: 10:30 AM, Weight: 88, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 8:13:00
ID: 17, Address: 3140 S 1100 W, Salt Lake City, UT, 84119, Deadline: EOD, Weight: 2, Status: En route, Departure Time: 9:33:00, Delivery Time: 10:15:40
ID: 18, Address: 1480 4800 S, Salt Lake City, UT, 84123, Deadline: EOD, Weight: 6, Status: At the hub, Departure Time: 11:00:00, Delivery Time: 12:25:40
ID: 19, Address: 177 W Price Ave, Salt Lake City, UT, 84115, Deadline: EOD, Weight: 37, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 8:49:40
ID: 20, Address: 3595 Main St, Salt Lake City, UT, 84115, Deadline: 10:30 AM, Weight: 37, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 8:48:00
ID: 21, Address: 3595 Main St, Salt Lake City, UT, 84115, Deadline: EOD, Weight: 3, Status: Delivered, Departure Time: 9:33:00, Delivery Time: 9:11:40
ID: 22, Address: 6351 South 900 East, Murray, UT, 84121, Deadline: EOD, Weight: 2, Status: En route, Departure Time: 9:33:00, Delivery Time: 10:41:00
ID: 23, Address: 5100 South 2700 West, Salt Lake City, UT, 84119, Deadline: EOD, Weight: 5, Status: En route, Departure Time: 9:33:00, Delivery Time: 11:05:00
ID: 24, Address: 5085 State St, Murray, UT, 84107, Deadline: EOD, Weight: 7, Status: En route, Departure Time: 9:33:00, Delivery Time: 10:31:00
ID: 25, Address: 5383 South 900 East #104, Salt Lake City, UT, 84117, Deadline: 10:30 AM, Weight: 7, Status: En route, Departure Time: 9:33:00, Delivery Time: 10:36:40
ID: 26, Address: 5383 South 900 East #104, Salt Lake City, UT, 84117, Deadline: EOD, Weight: 25, Status: At the hub, Departure Time: 11:00:00, Delivery Time: 11:08:00
ID: 27, Address: 1040 Dalton Ave S, Salt Lake City, UT, 84104, Deadline: EOD, Weight: 5, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 9:13:40
ID: 28, Address: 2835 Main St, Salt Lake City, UT, 84115, Deadline: EOD, Weight: 7, Status: At the hub, Departure Time: 11:00:00, Delivery Time: 11:27:40
ID: 29, Address: 1330 2100 S, Salt Lake City, UT, 84106, Deadline: 10:30 AM, Weight: 2, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 8:29:40
ID: 30, Address: 300 State St, Salt Lake City, UT, 84103, Deadline: 10:30 AM, Weight: 1, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 9:13:00
ID: 31, Address: 3345 S 900 W, Salt Lake City, UT, 84119, Deadline: 10:30 AM, Weight: 1, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 8:58:40
ID: 32, Address: 3345 S 900 W, Salt Lake City, UT, 84119, Deadline: EOD, Weight: 1, Status: At the hub, Departure Time: 11:00:00, Delivery Time: 11:36:40
ID: 33, Address: 2530 S 500 E, Salt Lake City, UT, 84106, Deadline: EOD, Weight: 1, Status: Delivered, Departure Time: 9:33:00, Delivery Time: 9:19:20
ID: 34, Address: 4500 S 2300 E, Holladay, UT, 84117, Deadline: 10:30 AM, Weight: 2, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 8:13:00
ID: 35, Address: 1840 Dalton Ave S, Salt Lake City, UT, 84104, Deadline: EOD, Weight: 88, Status: At the hub, Departure Time: 11:00:00, Delivery Time: 11:37:20
ID: 36, Address: 2300 Parkway Blvd, West Valley City, UT, 84119, Deadline: EOD, Weight: 88, Status: At the hub, Departure Time: 11:00:00, Delivery Time: 11:40:00
ID: 37, Address: 418 S State St, Salt Lake City, UT, 84111, Deadline: 10:30 AM, Weight: 2, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 9:29:40
ID: 38, Address: 418 S State St, Salt Lake City, UT, 84111, Deadline: EOD, Weight: 9, Status: At the hub, Departure Time: 11:00:00, Delivery Time: 12:13:20
ID: 39, Address: 2010 W 500 S, Salt Lake City, UT, 84104, Deadline: EOD, Weight: 9, Status: Delivered, Departure Time: 9:33:00, Delivery Time: 9:57:20

*****
WGUPS-950-Data-Structures-and-Algorithms-II - main.py 78/20 (46 chars) LF UTF-8 4 spaces Python 3.11 (WGUPS-Data-Structures-and-Algorithms-II)
```

D3. Third Status Check

Screenshot goes here



```
Project > main.py > packageCSV.csv > distanceCSV.csv > addressCSV.csv > hashtable.py > truck.py > package.py
# Helvin Bactigonog
# Student ID: 00991391
# C950 WGUPS ROUTING PROGRAM

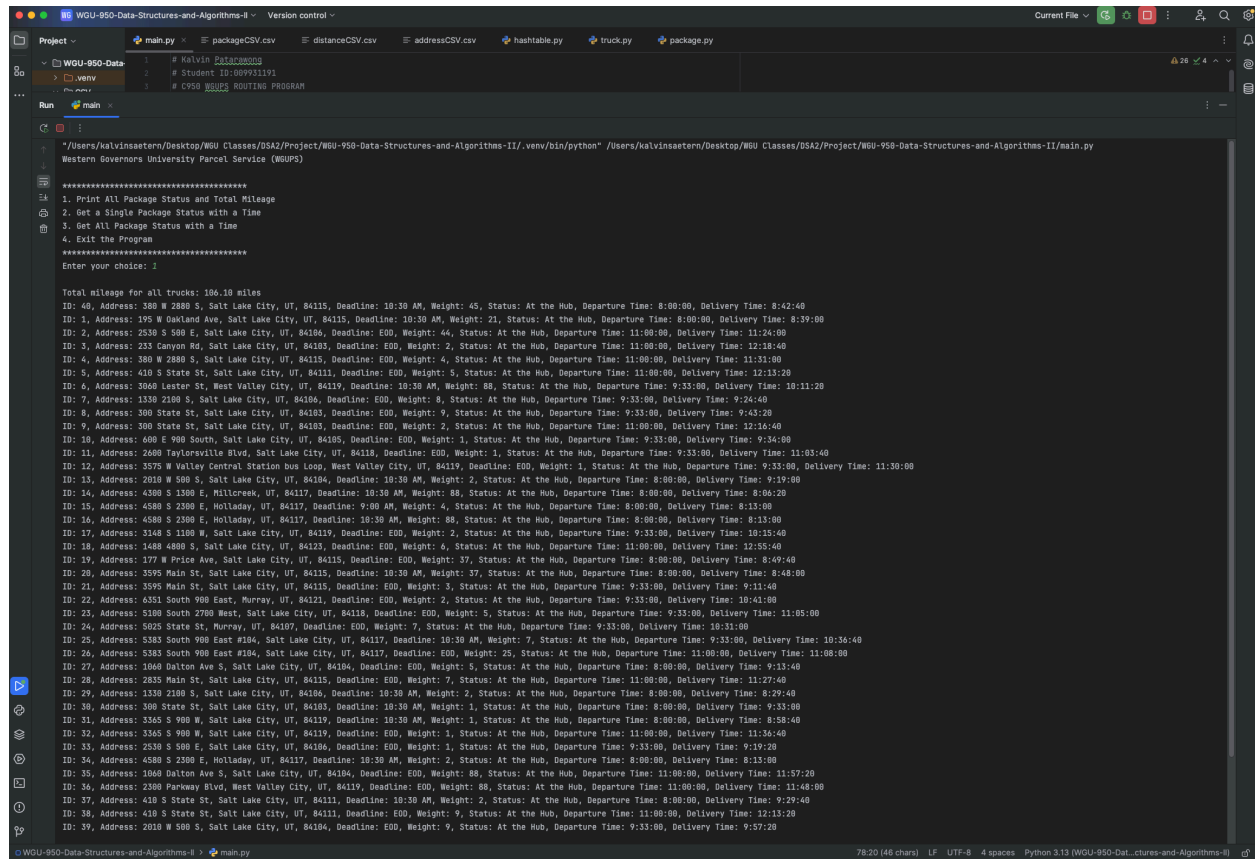
Run main
Western Governors University Parcel Service (WGUPS)

*****
1. Print All Package Status and Total Mileage
2. Get a Single Package Status with a Time
3. Get All Package Status with a Time
4. Exit the Program
*****
Enter Your Choice: 1
Enter a Time (MM:SS): 12:30
ID: 48, Address: 388 W 2880 S, Salt Lake City, UT, 84115, Deadline: 10:30 AM, Weight: 45, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 8:42:40
ID: 1, Address: 195 W Oakland Ave, Salt Lake City, UT, 84115, Deadline: 10:30 AM, Weight: 21, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 8:39:00
ID: 2, Address: 2530 S 500 E, Salt Lake City, UT, 84106, Deadline: EOD, Weight: 44, Status: Delivered, Departure Time: 11:00:00, Delivery Time: 11:24:00
ID: 3, Address: 233 Canyon Rd, Salt Lake City, UT, 84103, Deadline: EOD, Weight: 2, Status: Delivered, Departure Time: 11:00:00, Delivery Time: 12:18:40
ID: 4, Address: 388 W 2880 S, Salt Lake City, UT, 84115, Deadline: EOD, Weight: 4, Status: Delivered, Departure Time: 11:00:00, Delivery Time: 11:31:00
ID: 5, Address: 410 S State St, Salt Lake City, UT, 84111, Deadline: EOD, Weight: 5, Status: Delivered, Departure Time: 11:00:00, Delivery Time: 12:13:20
ID: 6, Address: 3648 Lester St, West Valley City, UT, 84119, Deadline: 10:30 AM, Weight: 88, Status: Delivered, Departure Time: 9:33:00, Delivery Time: 10:11:20
ID: 7, Address: 1330 2100 S, Salt Lake City, UT, 84106, Deadline: EOD, Weight: 8, Status: Delivered, Departure Time: 9:33:00, Delivery Time: 9:24:40
ID: 8, Address: 300 State St, Salt Lake City, UT, 84103, Deadline: EOD, Weight: 9, Status: Delivered, Departure Time: 9:33:00, Delivery Time: 9:43:20
ID: 9, Address: 410 S State St, Salt Lake City, UT, 84111, Deadline: EOD, Weight: 2, Status: Delivered, Departure Time: 11:00:00, Delivery Time: 12:16:40
ID: 10, Address: 408 E 900 South, Salt Lake City, UT, 84105, Deadline: EOD, Weight: 1, Status: Delivered, Departure Time: 9:33:00, Delivery Time: 9:34:00
ID: 11, Address: 2600 Taylorsville Blvd, Salt Lake City, UT, 84119, Deadline: EOD, Weight: 1, Status: Delivered, Departure Time: 9:33:00, Delivery Time: 11:03:40
ID: 12, Address: 3575 W Valley Central Station Bus Loop, West Valley City, UT, 84119, Deadline: EOD, Weight: 1, Status: Delivered, Departure Time: 9:33:00, Delivery Time: 11:30:00
ID: 13, Address: 2010 W 500 S, Salt Lake City, UT, 84104, Deadline: 10:30 AM, Weight: 2, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 9:19:00
ID: 14, Address: 4300 S 1300 E, Hillcrest, UT, 84117, Deadline: 10:30 AM, Weight: 88, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 8:06:20
ID: 15, Address: 4580 S 2300 E, Holladay, UT, 84117, Deadline: 9:00 AM, Weight: 4, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 8:13:00
ID: 16, Address: 4580 S 2300 E, Holladay, UT, 84117, Deadline: 10:30 AM, Weight: 80, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 8:13:00
ID: 17, Address: 3140 S 1100 W, Salt Lake City, UT, 84119, Deadline: EOD, Weight: 2, Status: Delivered, Departure Time: 9:33:00, Delivery Time: 10:15:40
ID: 18, Address: 1408 4800 S, Salt Lake City, UT, 84122, Deadline: EOD, Weight: 6, Status: En route, Departure Time: 11:00:00, Delivery Time: 12:55:40
ID: 19, Address: 177 W Price Ave, Salt Lake City, UT, 84115, Deadline: EOD, Weight: 37, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 8:49:40
ID: 20, Address: 3595 Main St, Salt Lake City, UT, 84115, Deadline: 10:30 AM, Weight: 37, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 8:48:00
ID: 21, Address: 3595 Main St, Salt Lake City, UT, 84115, Deadline: EOD, Weight: 3, Status: Delivered, Departure Time: 9:33:00, Delivery Time: 9:11:40
ID: 22, Address: 4351 South 900 East, Murray, UT, 84121, Deadline: EOD, Weight: 2, Status: Delivered, Departure Time: 9:33:00, Delivery Time: 10:41:00
ID: 23, Address: 5100 South 2700 West, Salt Lake City, UT, 84118, Deadline: EOD, Weight: 5, Status: Delivered, Departure Time: 9:33:00, Delivery Time: 11:05:00
ID: 24, Address: 5025 State St, Murray, UT, 84107, Deadline: EOD, Weight: 7, Status: Delivered, Departure Time: 9:33:00, Delivery Time: 10:31:00
ID: 25, Address: 5383 South 900 East #104, Salt Lake City, UT, 84117, Deadline: 10:30 AM, Weight: 7, Status: Delivered, Departure Time: 9:33:00, Delivery Time: 10:36:40
ID: 26, Address: 5383 South 900 East #104, Salt Lake City, UT, 84117, Deadline: EOD, Weight: 25, Status: Delivered, Departure Time: 11:00:00, Delivery Time: 11:08:00
ID: 27, Address: 1060 Dalton Ave S, Salt Lake City, UT, 84104, Deadline: EOD, Weight: 5, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 9:13:40
ID: 28, Address: 2835 Main St, Salt Lake City, UT, 84115, Deadline: EOD, Weight: 7, Status: Delivered, Departure Time: 11:00:00, Delivery Time: 11:27:40
ID: 29, Address: 1350 2100 S, Salt Lake City, UT, 84106, Deadline: 10:30 AM, Weight: 2, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 8:29:40
ID: 30, Address: 300 State St, Salt Lake City, UT, 84103, Deadline: 10:30 AM, Weight: 1, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 9:33:00
ID: 31, Address: 3365 S 900 W, Salt Lake City, UT, 84119, Deadline: 10:30 AM, Weight: 1, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 8:58:40
ID: 32, Address: 3365 S 900 W, Salt Lake City, UT, 84119, Deadline: EOD, Weight: 1, Status: Delivered, Departure Time: 11:00:00, Delivery Time: 11:36:40
ID: 33, Address: 2530 S 500 E, Salt Lake City, UT, 84106, Deadline: EOD, Weight: 1, Status: Delivered, Departure Time: 9:33:00, Delivery Time: 9:19:20
ID: 34, Address: 4580 S 2300 E, Holladay, UT, 84117, Deadline: 10:30 AM, Weight: 2, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 8:13:00
ID: 35, Address: 1000 Dalton Ave S, Salt Lake City, UT, 84104, Deadline: EOD, Weight: 80, Status: Delivered, Departure Time: 11:00:00, Delivery Time: 11:37:00
ID: 36, Address: 2300 Parkway Blvd, West Valley City, UT, 84119, Deadline: EOD, Weight: 80, Status: Delivered, Departure Time: 11:00:00, Delivery Time: 11:48:00
ID: 37, Address: 410 S State St, Salt Lake City, UT, 84111, Deadline: 10:30 AM, Weight: 2, Status: Delivered, Departure Time: 8:00:00, Delivery Time: 9:29:40
ID: 38, Address: 410 S State St, Salt Lake City, UT, 84111, Deadline: EOD, Weight: 9, Status: Delivered, Departure Time: 11:00:00, Delivery Time: 12:13:20
ID: 39, Address: 2010 W 500 S, Salt Lake City, UT, 84104, Deadline: EOD, Weight: 9, Status: Delivered, Departure Time: 9:33:00, Delivery Time: 9:57:20

*****
```

E. Screenshot of Code Execution

Screenshot goes here



```
#!/usr/bin/env python3
# Package Delivery Problem
# Student ID: 009231191
# C950 WGUPS ROUTING PROGRAM

# Print All Package Status and Total Mileage
# Get a Single Package Status with a Time
# Get All Package Status with a Time
# Exit the Program

Enter your choice: 1

Total mileage for all trucks: 106.10 miles

ID: 48, Address: 380 W 2880 S, Salt Lake City, UT, 84115, Deadline: 10:30 AM, Weight: 45, Status: At the Hub, Departure Time: 8:00:00, Delivery Time: 8:42:40
ID: 1, Address: 195 W Oakland Ave, Salt Lake City, UT, 84115, Deadline: 10:30 AM, Weight: 21, Status: At the Hub, Departure Time: 8:00:00, Delivery Time: 8:39:00
ID: 2, Address: 2530 S 500 E, Salt Lake City, UT, 84106, Deadline: EOD, Weight: 44, Status: At the Hub, Departure Time: 11:00:00, Delivery Time: 11:24:00
ID: 3, Address: 253 Canyon Rd, Salt Lake City, UT, 84103, Deadline: EOD, Weight: 2, Status: At the Hub, Departure Time: 11:00:00, Delivery Time: 12:18:40
ID: 4, Address: 380 W 2880 S, Salt Lake City, UT, 84115, Deadline: EOD, Weight: 4, Status: At the Hub, Departure Time: 11:00:00, Delivery Time: 11:31:00
ID: 5, Address: 410 S State St, Salt Lake City, UT, 84111, Deadline: EOD, Weight: 5, Status: At the Hub, Departure Time: 11:00:00, Delivery Time: 12:13:20
ID: 6, Address: 3060 Lester St, West Valley City, UT, 84119, Deadline: 10:30 AM, Weight: 88, Status: At the Hub, Departure Time: 9:33:00, Delivery Time: 10:11:20
ID: 7, Address: 1330 2100 S, Salt Lake City, UT, 84106, Deadline: EOD, Weight: 8, Status: At the Hub, Departure Time: 9:33:00, Delivery Time: 9:24:40
ID: 8, Address: 300 State St, Salt Lake City, UT, 84103, Deadline: EOD, Weight: 9, Status: At the Hub, Departure Time: 9:33:00, Delivery Time: 9:43:20
ID: 9, Address: 300 State St, Salt Lake City, UT, 84103, Deadline: EOD, Weight: 2, Status: At the Hub, Departure Time: 11:00:00, Delivery Time: 12:16:40
ID: 10, Address: 400 E 900 South, Salt Lake City, UT, 84105, Deadline: EOD, Weight: 1, Status: At the Hub, Departure Time: 9:33:00, Delivery Time: 9:34:00
ID: 11, Address: 2600 Taylorsville Blvd, Salt Lake City, UT, 84118, Deadline: EOD, Weight: 1, Status: At the Hub, Departure Time: 9:33:00, Delivery Time: 11:03:40
ID: 12, Address: 3575 W Valley Central Station bus Loop, West Valley City, UT, 84119, Deadline: EOD, Weight: 1, Status: At the Hub, Departure Time: 9:33:00, Delivery Time: 11:30:00
ID: 13, Address: 2010 W 500 S, Salt Lake City, UT, 84106, Deadline: 10:30 AM, Weight: 2, Status: At the Hub, Departure Time: 8:00:00, Delivery Time: 9:19:00
ID: 14, Address: 4300 S 1300 E, Hillcrest, UT, 84117, Deadline: 10:30 AM, Weight: 80, Status: At the Hub, Departure Time: 8:00:00, Delivery Time: 8:06:20
ID: 15, Address: 4580 S 2300 E, Holladay, UT, 84117, Deadline: 9:00 AM, Weight: 4, Status: At the Hub, Departure Time: 8:00:00, Delivery Time: 8:13:00
ID: 16, Address: 4580 S 2300 E, Holladay, UT, 84117, Deadline: 10:30 AM, Weight: 88, Status: At the Hub, Departure Time: 8:00:00, Delivery Time: 8:13:00
ID: 17, Address: 3148 S 1100 W, Salt Lake City, UT, 84119, Deadline: EOD, Weight: 2, Status: At the Hub, Departure Time: 9:33:00, Delivery Time: 10:15:40
ID: 18, Address: 1488 4800 S, Salt Lake City, UT, 84123, Deadline: EOD, Weight: 6, Status: At the Hub, Departure Time: 11:00:00, Delivery Time: 12:55:40
ID: 19, Address: 177 W Price Ave, Salt Lake City, UT, 84115, Deadline: EOD, Weight: 37, Status: At the Hub, Departure Time: 8:00:00, Delivery Time: 8:49:40
ID: 20, Address: 3595 Main St, Salt Lake City, UT, 84115, Deadline: 10:30 AM, Weight: 37, Status: At the Hub, Departure Time: 8:00:00, Delivery Time: 8:48:00
ID: 21, Address: 3595 Main St, Salt Lake City, UT, 84115, Deadline: EOD, Weight: 3, Status: At the Hub, Departure Time: 9:33:00, Delivery Time: 9:11:40
ID: 22, Address: 6351 South 900 East, Murray, UT, 84121, Deadline: EOD, Weight: 2, Status: At the Hub, Departure Time: 9:33:00, Delivery Time: 10:41:00
ID: 23, Address: 5100 South 2700 West, Salt Lake City, UT, 84118, Deadline: EOD, Weight: 5, Status: At the Hub, Departure Time: 9:33:00, Delivery Time: 11:05:00
ID: 24, Address: 5025 State St, Murray, UT, 84107, Deadline: EOD, Weight: 7, Status: At the Hub, Departure Time: 9:33:00, Delivery Time: 10:31:00
ID: 25, Address: 5383 South 900 East #104, Salt Lake City, UT, 84117, Deadline: 10:30 AM, Weight: 7, Status: At the Hub, Departure Time: 9:33:00, Delivery Time: 10:36:40
ID: 26, Address: 5383 South 900 East #104, Salt Lake City, UT, 84117, Deadline: EOD, Weight: 25, Status: At the Hub, Departure Time: 11:00:00, Delivery Time: 11:08:00
ID: 27, Address: 1040 Dalton Ave S, Salt Lake City, UT, 84104, Deadline: EOD, Weight: 5, Status: At the Hub, Departure Time: 8:00:00, Delivery Time: 9:13:40
ID: 28, Address: 2835 Main St, Salt Lake City, UT, 84115, Deadline: EOD, Weight: 7, Status: At the Hub, Departure Time: 11:00:00, Delivery Time: 11:27:40
ID: 29, Address: 1330 2100 S, Salt Lake City, UT, 84106, Deadline: 10:30 AM, Weight: 2, Status: At the Hub, Departure Time: 8:00:00, Delivery Time: 8:29:40
ID: 30, Address: 300 State St, Salt Lake City, UT, 84103, Deadline: 10:30 AM, Weight: 1, Status: At the Hub, Departure Time: 8:00:00, Delivery Time: 9:33:00
ID: 31, Address: 3365 S 900 W, Salt Lake City, UT, 84119, Deadline: 10:30 AM, Weight: 1, Status: At the Hub, Departure Time: 8:00:00, Delivery Time: 8:58:40
ID: 32, Address: 3365 S 900 W, Salt Lake City, UT, 84119, Deadline: EOD, Weight: 1, Status: At the Hub, Departure Time: 11:00:00, Delivery Time: 11:35:40
ID: 33, Address: 2530 S 500 E, Salt Lake City, UT, 84106, Deadline: EOD, Weight: 1, Status: At the Hub, Departure Time: 9:33:00, Delivery Time: 9:13:20
ID: 34, Address: 4580 S 2300 E, Holladay, UT, 84117, Deadline: 10:30 AM, Weight: 2, Status: At the Hub, Departure Time: 8:00:00, Delivery Time: 8:13:00
ID: 35, Address: 1040 Dalton Ave S, Salt Lake City, UT, 84104, Deadline: EOD, Weight: 88, Status: At the Hub, Departure Time: 11:00:00, Delivery Time: 11:57:20
ID: 36, Address: 2300 Parkway Blvd, West Valley City, UT, 84119, Deadline: EOD, Weight: 88, Status: At the Hub, Departure Time: 11:00:00, Delivery Time: 11:48:00
ID: 37, Address: 410 S State St, Salt Lake City, UT, 84111, Deadline: 10:30 AM, Weight: 2, Status: At the Hub, Departure Time: 8:00:00, Delivery Time: 9:29:40
ID: 38, Address: 410 S State St, Salt Lake City, UT, 84111, Deadline: EOD, Weight: 9, Status: At the Hub, Departure Time: 11:00:00, Delivery Time: 12:13:20
ID: 39, Address: 2010 W 500 S, Salt Lake City, UT, 84106, Deadline: EOD, Weight: 9, Status: At the Hub, Departure Time: 9:33:00, Delivery Time: 9:27:20
```

F1. Strengths of the Chosen Algorithm

The package delivery algorithm used in the solution, based on the Nearest Neighbor Algorithm, has multiple strengths. It is efficient, offering a quick solution to route planning and reducing complexity. It is also simple and easy to understand and maintain, making it ideal for smaller-scale scenarios. The algorithm can handle multiple classes and works well with non-linear data, ensuring reliability even when the path between delivery points is unclear. These strengths make it a practical choice for the delivery problem in the original program.

F2. Verification of Algorithm

The Nearest Neighbor Algorithm effectively meets all the requirements for this scenario by ensuring all packages are delivered on time, keeping the total mileage under 140 miles, and accommodating special delivery instructions. Packages that couldn't depart until 9:05 were scheduled accordingly, and those that needed to be delivered together were handled on the same trip. The algorithm guarantees that no package is left behind, fulfilling the delivery process's time and logistical needs.

F3. Other Possible Algorithms

Two other algorithms that could meet the requirements in this scenario are Dynamic Programming and Genetic Algorithm. Dynamic Programming could solve the Traveling Salesman Problem with time windows, ensuring an optimal solution, though with higher computational complexity. Genetic Algorithms, on the other hand, could optimize routes by evolving solutions over multiple generations, considering constraints like time windows, deadlines, and total mileage.

F3a. Algorithm Differences

Dynamic Programming differs from the algorithm used in the solution because it guarantees an optimal route but comes with a higher computational cost. DP evaluates all possible routes and eliminates non-optimal ones, which can take a significant amount of time.

On the other hand, the Genetic Algorithm uses a probabilistic approach to evolve solutions over multiple generations. While it doesn't always guarantee the optimal solution, it can find a good one more quickly, making it a more efficient option in terms of computational time.

G. Different Approach

If I were to do this project again, I'd make a few changes to improve it. First, I'd add a feature to prioritize packages based on their deadlines or special instructions, ensuring those are delivered first. I'd also refactor the code to sort packages more efficiently, making the delivery process smoother. Finally, I'd work on making the user interface more interactive and user-friendly so it's easier to navigate and use.

H. Verification of Data Structure

The hash table used in the solution meets the scenario's requirements by efficiently storing and accessing package data. With each package having a unique ID, the hash table allows for quick lookups, which is vital for retrieving package details during delivery. This speed is essential for the algorithm to work efficiently. The hash table also offers flexibility-packages can be added or removed as needed when trucks are loaded, and their details can be easily updated. This dynamic functionality ensures that the program can adapt to changes in the delivery process while maintaining accuracy and performance.

H1. Other Data Structures

Two other data structures that could meet the requirements are a doubly linked list, which allows efficient insertion and deletion of packages with bidirectional traversal, and a queue, which works well for managing packages in a first-in-first-out order for sequential processing.

H1a. Data Structure Differences

The doubly linked list differs from the hash table because it stores elements in a sequence with pointers to the previous and next nodes, making inserting and deleting packages efficient. However, searching for a specific package takes longer, with an $O(n)$ time complexity. In contrast, a hash table allows for constant-time lookups $O(1)$ by using the unique package ID as the key, making it much faster for searching and retrieving data.

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On the other hand, the queue processes data in a first-in-first-out (FIFO) order, which works well for sequentially handling packages. However, this structure isn't as flexible when dealing with packages that need to be accessed or delivered out of sequence, as it processes them in a strict order. The hash table, by contrast, gives direct access to packages based on their ID, making it better suited for handling more complex delivery requirements like deadlines or specific routes.

I. Sources

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